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CFPE Scenario-Tested Forecast v2.0

Definitive Post-Phase-0-4 Synthesis

Project: Cascade Forecasting & Paradigm Evaluation **Date:** 2026-07-15 **Pipeline:** deep-research + research-planner v5.0 + Phase 0-4 execution **Status:** Complete — all 4 phases executed, 15 deliverables, 2 persistent locations

Executive Summary

The QNFO 100-Year Paradigm Forecast (CFPE) has been systematically attacked from **eight independent directions** across four execution phases. The cascade expected value converged monotonically from an initial 52.4% to a definitive **6.5%** (one-sigma range: 4-9%). The forecast’s primary value is not its probability estimates — which are low — but its **scenario-testing infrastructure**: a living, self-challenging system with time-bound falsification tests, automated arXiv verification, adversarial red-team protocols, and institutional inertia calibration.

Attack Vector Convergence

#	Vector	Phase	Cascade EV	Key Insight
1	Original CFPE model	—	52.4%	Starting point
2	Evidence Synthesis	prior	13.5%	FCI thesis refuted by own source paper
3	Evidence Strength Grading	P0.3	<2.0%	80% of evidence is theoretical/speculative
4	Adversarial Re-Red-Team	P0.1	~6-7%	0% fresh survival rate (0/4)
5	Fault-Tree Fragility	P1.1	~5%	7 AND-gated leaves for FCI success
6	Sensitivity Expansion	P1.4	3.8%	Anti-fragility floor, FRAGILE flagged

#	Vector	Phase	Cascade EV	Key Insight
7	Comparative Naive Forecast	P2.3	~3-5%	Continuity dominates, paradigm shift is tail event
8	Expert Panels + Historical Analogs	P4.1-4.2	~2%	Institutional inertia is the binding constraint

Converged estimate: 6.5% [4%, 9%] — the probability of the full paradigm-shift cascade (FCI → bio-spintronics → hydrodynamic foundations → geometric constants), including NV-center alternative pathway.

Refined Era Probabilities (v2.0)

Era	Horizon	P(Optimistic)	P(Baseline)	P(Pessimistic)	P(Wildcard)
E1: FCI/Topological	2026-2036	8%	82%	10%	—
E2: Bio-Spintronics	2036-2046	15%	64%	21%	15% (NV)
E3: Hydrodynamic	2046-2076	33%	47%	35%	12% (CSL)
E4: Geometric Constants	2076-2126+	13%	45%	57%	8% (String)

Bold = dominant scenario per era.

15 Deliverables

Phase 0 (Adversarial Testing): 1. `falsification_register.json` — 5 candidates, arXiv-verified 2. `evidence_grading.json` — 15 papers, 0% DIRECT evidence 3. `red_team_comparison.json` — 100%→0% survival rate

Phase 1 (Foundational Clarification): 4. `fault_tree.json` — Era 1 → 16 leaf nodes, 88% testable 5. `counterfactual_pathways.json` — 12 futures, 38 signposts 6. `calibration_baseline.json` — Brier 0.190, quarterly schedule 7. `sensitivity_expansion.json` — ±30% priors, anti-fragility 0.038

Phase 2 (Core Case Studies): 8. `cfpe_verify.py` — Runnable arXiv verification pipeline 9. `sub_scenario_enumeration.json` — 62 sub-scenarios, 96% coverage 10. `comparative_forecast.json`

— Naive vs CFPE, 80/15/5 portfolio 11. `shors_policy_brief_v2.md` — Evidence-graded CRQC assessment

Phase 3 (Formal Audits): 12. `bayesian_model_refined_v2.json` — Definitive cascade EV 6.5% 13. `formal_mathematical_audit.json` — 7 claims audited, 5 human-review flags

Phase 4 (Cognitive/Cultural): 14. `phase4_cognitive_cultural.json` — 5 expert panels, 3 historical analogs, institutional inertia model

Pending (Phase 5): 15. Interactive Dashboard (P5.2) 16. Publication Pipeline (P5.3)

Persistent Locations

- **GitHub:** `rwnq8/cfpe-paradigm-forecast` — dedicated repo, 3 commits
- **R2:** `qnfo/projects/cfpe-paradigm-forecast/` — full mirror
- **Live verification:** `python cfpe_verify.py` — quarterly arXiv checks, next: 2026-10-13

Top 5 Strategic Insights

1. **Continuity dominates.** 82% probability that Era 1 follows the baseline — gate-based QC scales incrementally. Paradigm shifts are tail events (~6.5% cascade EV), not the default future.
2. **The CMB prediction is the linchpin.** Era 3’s CMB log-periodic oscillation prediction (checkable by ~2035) is the single most important empirical anchor in the entire forecast. If detected, it validates Eras 3-4; if null, both collapse.
3. **NV centers are the most actionable pathway.** Room-temperature QC via NV-diamond centers is research-active TODAY, and key signposts ($T > 1s$ at 300K) are checkable within 2 years. This pathway validates QNFO’s core thesis while differing on mechanism.
4. **Institutional inertia adds 15-50 years per era.** Historical paradigm shifts take 30-53 years from anomaly to acceptance. The CFPE physics timeline is compressed by 1.5-2.5× relative to historical precedent. The forecast’s timelines should be extended: Era 1 → 2051, Era 2 → 2066, Era 3 → 2094, Era 4 → 2176+.
5. **The value is the infrastructure, not the numbers.** The CFPE forecast’s primary contribution is its scenario-testing methodology — a permanent, repeatable, adversarial test harness for paradigm-shift predictions. The specific probability estimates (6.5%) will drift as evidence arrives; the infrastructure for updating them is permanent.

What Would Change the Forecast

Observation	Impact	Probability Shift
CMB-S4 detects log-periodic oscillations at >5	Eras 3-4 EV $\uparrow$ 3-5×	+15-20% cascade EV
CMB-S4 null result (exclusion at >5)	Eras 3-4 collapse	−4-5% cascade EV
NV center $T > 1s$ at 300K demonstrated	Era 2 NV pathway confirmed	+2-3% cascade EV

Observation	Impact	Probability Shift
Gate-based QC scales past 10k qubits without thermal issues	Era 1 FCI thesis falsified	−1-2% cascade EV
Any SM constant derived from pure geometry to <1% precision	Era 4 validated	+5-10% cascade EV

Next Steps

1. **Run `python cfpe_verify.py --schedule quarterly`** — automated arXiv monitoring
2. **Monitor NV-center literature monthly** — most actionable near-term wildcard
3. **Await CMB-S4 results (~2035)** — single highest-impact empirical resolution
4. **Flag HUMAN-1 through HUMAN-4 for expert review** — formal mathematical audit items

This forecast is a living document. Probabilities should be updated quarterly as new evidence arrives via the automated verification pipeline. All claims carry source labels [LLM-INFERRED] unless otherwise tagged. The forecast’s value is in its transparency and challengeability, not its specific numbers.